

Pennada Venkata Chaitanya

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Professional Summary

AI/ML Software Engineer with **3+ years of hands-on experience** building production Generative AI systems, AI agents, RAG applications, distributed backend services, and scalable data pipelines. Experienced in Python, FastAPI, LLMs, LangGraph, MCP, vector search, PostgreSQL, MongoDB, and production AI automation.

Professional Experience

Software Engineer | Innocito

April 2025 – Present

- Built **distributed data pipelines** using Python, Prefect, FastAPI, and MongoDB, leveraging asynchronous and concurrent processing to automate production workflows and reduce processing time from **14 days to 1 day**.
- Processed and mapped **5 million+ product records** through high-throughput AI pipelines for large-scale data preprocessing, classification, and catalog enrichment.
- Built a real-time **AI Voice Agent** integrating LLMs, speech-to-text, and text-to-speech, reducing end-to-end response latency by **3 seconds per request** and enabling more natural conversations.
- Implemented local and remote **Model Context Protocol (MCP)** servers for structured tool routing and integration between AI agents and backend services.

AI Software Developer | AI Gen media

March 2024 – February 2025

- Developed AI-driven text-to-video workflows, achieving **80% visual consistency** across predefined test cases.
- Built an automated AI GIF Studio, achieving **95% test-case accuracy**, **5× faster runs**, and **10,000+ interactions**.
- Developed AI agents for social media automation, reducing manual task time by **30%**.

SDE AI/ML Intern | Spring Works

July 2023 – September 2023

- Increased chatbot test coverage from **30% to 75%** using Python and Pytest-based automated testing.
- Collaborated on developing and deploying **Generative AI Proofs of Concept**.

AI/ML Intern | Technical Hub

July 2022 – November 2022

- Built and evaluated machine-learning models using Python, performing data preprocessing, feature engineering, and model validation to improve prediction performance.

Projects

Synapse

[Website](#) | [GitHub](#)

- Architected an **offline-first AI knowledge system** combining block-based knowledge management, information retrieval, task orchestration, and agent-driven actions.
- Designed **distributed synchronization** between SQLite and PostgreSQL using block-level incremental updates and Hybrid Logical Clock (HLC)-based conflict resolution.
- Engineered **hybrid local-cloud Retrieval-Augmented Generation (RAG)** pipelines with embedding generation, incremental vector indexing, semantic retrieval, context assembly, and citation-grounded LLM response generation.
- Integrated **LangGraph-based agent orchestration** with MCP tool execution and local/cloud LLM workflows, enabling retrieved knowledge to drive structured actions while supporting local-first AI execution.

Tech: Python, FastAPI, PostgreSQL, SQLite, LangGraph, Ollama, MCP, Vector Search.

Ed AI

[GitHub](#)

- Developed an **AI-powered technical learning platform** with context-aware Socratic tutoring, personalized recommendations, coding practice, and AI-driven mock interviews.
- Engineered a **RAG tutoring pipeline** grounding LLM responses in course content, learner progress, and weak areas.

Tech: Python, FastAPI, Gemini, Chroma, RAG, Vector Search.

Skills

Programming: Python, SQL, C++

Generative AI & Agents: Generative AI, LLMs, AI Agents, RAG, Multimodal LLMs, LangGraph, LangChain, LlamaIndex, MCP, Vector Search, Ollama

Machine Learning & AI: PyTorch, NLP, Computer Vision, Model Evaluation, Data Preprocessing, Feature Engineering

Backend & Data: FastAPI, REST APIs, PostgreSQL, MongoDB (NoSQL), SQLite, Prefect, Asynchronous Processing

Engineering Tools: Docker, Pytest, Playwright, Git, GitHub, Jira, Agile/Scrum, Automation, Streamlit

Certifications

- **Azure AI Fundamentals** – Microsoft
- **The Joy of Computing Using Python** – NPTEL
- **Data Science for Beginners** – NASSCOM
- **MTA Python Certification** – Microsoft

Achievements

- Won the institutional round of the **Solving for India Hackathon** hosted by Google, AMD, and GeeksforGeeks. [Certificate](#)
- Won the international **Fresh Hacks** hackathon hosted on [Devpost](#).
- Received the **Rising Star Award** at Innocito for strong performance, rapid growth, and technical contributions.
- Earned a **5-star Problem Solving** rating on [HackerRank](#) and solved **300+ DSA problems** across [LeetCode](#) and [GFG](#).

Volunteer Experience

SDE Mentor | Innovative Origin

April 2024 – March 2025

- Mentored **100+ students** in AI and Machine Learning, focusing on industry-relevant skills, practical applications, and software development guidance.
- Conducted interactive and personalized mentoring sessions, contributing **300+ volunteer hours** to the community.

Education

B.Tech in Computer Science | Aditya College of Engineering and Technology

2020 – 2024

Focus: Artificial Intelligence & Machine Learning